



ISO 9001:2015 Certified
Level I Institutionally Accredited

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

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BSIT 4A

DevCore Innovations

1.2 Nature of Business

DevCore Innovations is a newly established software development company that specializes in developing custom web applications, business software, mobile applications, and technology solutions for small and medium-sized businesses.

The company operates from a single office floor with 20 employees distributed across Information Technology, Human Resources, Finance, and Sales departments.

1.3 Company Vision

To become a trusted and innovative software development company that provides reliable, secure, and accessible technology solutions to businesses.

1.4 Company Mission

DevCore Innovations aims to deliver high-quality software solutions through innovation, collaboration, secure technology practices, and excellent customer service while providing employees with a productive and reliable IT environment.

1.5 Office Location

DevCore Innovations
2nd Floor, Innovation Center
123 Technology Avenue
Imus, Cavite, Philippines

Note: The address is fictional and used only for this project.

1.6 Organizational Structure

PART 1.6 — DevCore Innovations Organizational Structure

Management roles are included within the 20 departmental employees.

Position / Department	Employees	Main Responsibility
Chief Executive Officer (CEO)	1	Provides overall leadership and strategic direction
IT Manager	1	Manages IT infrastructure, systems, security, and IT staff
IT Department	5	Software development, systems administration, technical support, and infrastructure
HR Manager / HR Department	4	Employee records, recruitment, benefits, and HR operations
Finance Manager / Finance Department	5	Budgeting, accounting, financial records, and reporting
Sales Manager / Sales Department	6	Customer relations, sales activities, marketing, and business development
TOTAL COMPANY EMPLOYEES	20	IT (5) + HR (4) + Finance (5) + Sales (6)

1.7 Employee Distribution

PART 1 — Employee Distribution

Department	Employees	Primary IT Requirements
Information Technology	5	Development PCs, server access, Git, virtualization
Human Resources	4	Office applications, HR files, email, printing
Finance	5	Office applications, financial documents
Sales	6	Office applications, browser, CRM/web applications
TOTAL	20	

PART 2 — Enterprise Hardware Inventory

PART 2 — Enterprise Hardware Inventory				
Asset ID	Hardware	Qty.	Department	Purpose
PC-001-020	Desktop Computer	20	All Departments	Primary workstation
LAP-001-003	Business Laptop	3	IT / Management	Meetings, presentations, troubleshooting
SRV-001	Server	1	IT	File services, authentication, applications
RTR-001	Business Router	1	IT	Routing and internet connectivity
SW-001	24-Port Managed Switch	1	IT	Connects wired network devices
PRN-001	Network Laser Printer	1	All Departments	Shared office printing
UPS-001	UPS 1500VA	1	Server Room	Protects server equipment
UPS-002	UPS 1500VA	1	Server Room	Protects network equipment
AP-001-002	Wireless Access Point	2	Office	Wireless network coverage
NAS-001	4-Bay NAS	1	IT	Centralized storage and backups
BAK-001-002	External Backup Drive	2	IT	Offline backup and recovery
MON-001-020	24-inch Monitor	20	All Departments	Display for workstations
KBD-001-020	Keyboard	20	All Departments	User input
MSE-001-020	Mouse	20	All Departments	User input

The 20 desktop computers provide one dedicated workstation for each employee.

The 3 laptops are intended for IT personnel and management for meetings, presentations, troubleshooting, and mobility.

One server is sufficient for the company's initial 20-person environment. The server can provide centralized services while allowing future expansion.

The 24-port managed switch supports the current infrastructure while leaving additional ports available for future devices.

Two wireless access points provide better wireless coverage throughout the office.

The NAS and external backup drives provide additional protection against data loss and hardware failure.

PART 3 — Enterprise Software Inventory

PART 3 — Enterprise Software Inventory

Software	Version	License	Purpose
Windows 11 Pro	24H2	Commercial	Primary employee operating system
Ubuntu Server	24.04 LTS	Open Source	Server operating system
Microsoft 365 Apps	Current	Subscription	Productivity applications
Visual Studio Code	Current	Free	Software development
Git	Current	Open Source	Version control
GitHub Desktop	Current	Free	Graphical Git management
VirtualBox	Current	Open Source	Virtual machine testing
Google Chrome	Current	Free	Web browsing and web applications
Microsoft Defender	Included with Windows	Included	Endpoint security
AnyDesk	Current	Commercial / Free depending on use	Authorized remote support
7-Zip	Current	Open Source	File compression and extraction

Software Explanation

Windows 11 Pro will be used as the main operating system for employee computers because it provides a familiar business environment and supports professional management and security features.

Ubuntu Server 24.04 LTS will be used for server infrastructure because it is a stable long-term-support Linux platform.

Microsoft 365 Apps provides productivity applications required for documents, spreadsheets, presentations, and business work.

The IT department will use Visual Studio Code, Git, GitHub Desktop, and VirtualBox for software development, source-code management, and testing.

Google Chrome provides access to web applications and online services.

Microsoft Defender provides built-in malware and endpoint protection for Windows computers.

AnyDesk can be used by authorized IT personnel for remote technical support.

7-Zip provides file compression and extraction capabilities.

PART 4 — Enterprise Network Inventory

PART 4 — Enterprise Network Inventory				
Asset ID	Network Equipment	Qty.	Specification	Purpose
NET-001	ISP Modem/ONT	1	Business fiber modem/ONT	Internet termination
NET-002	Business Router	1	Gigabit / VLAN capable	Routing and network gateway
NET-003	Firewall	1	Business / Next-Generation Firewall	Network security
NET-004	Managed Switch	1	24-Port Gigabit	Wired connectivity
NET-005-006	Wireless Access Point	2	Wi-Fi 6	Wireless connectivity
NET-007	Patch Panel	1	24-Port CAT6	Cable organization
CAB-001-040	CAT6 Ethernet Cable	40	CAT6	Network connections
CON-001-100	RJ45 Connectors	100	CAT6 compatible	Cable termination
RCK-001	Network Rack	1	12U or larger	Equipment housing
PDU-001	Rack PDU	1	Standard rack PDU	Power distribution

Proposed VLAN Structure

PART 4 — Proposed VLAN Structure		
VLAN	Name	Purpose
10	IT	IT Department
20	HR	Human Resources
30	Finance	Finance Department
40	Sales	Sales Department
50	Servers	Server Infrastructure
60	Guest	Guest Wi-Fi
99	Management	Network Device Administration

VLAN segmentation improves security by separating departments and limiting unnecessary network communication.

PART 5 — Enterprise Network Diagram

Diagram Title

DevCore Innovations — Enterprise Network Topology

Student Name: [Student Name]

The Draw.io diagram should contain:

Internet

ISP Modem

Router

Firewall

Managed Switch

Wireless Access Point

Server

Printer

NAS

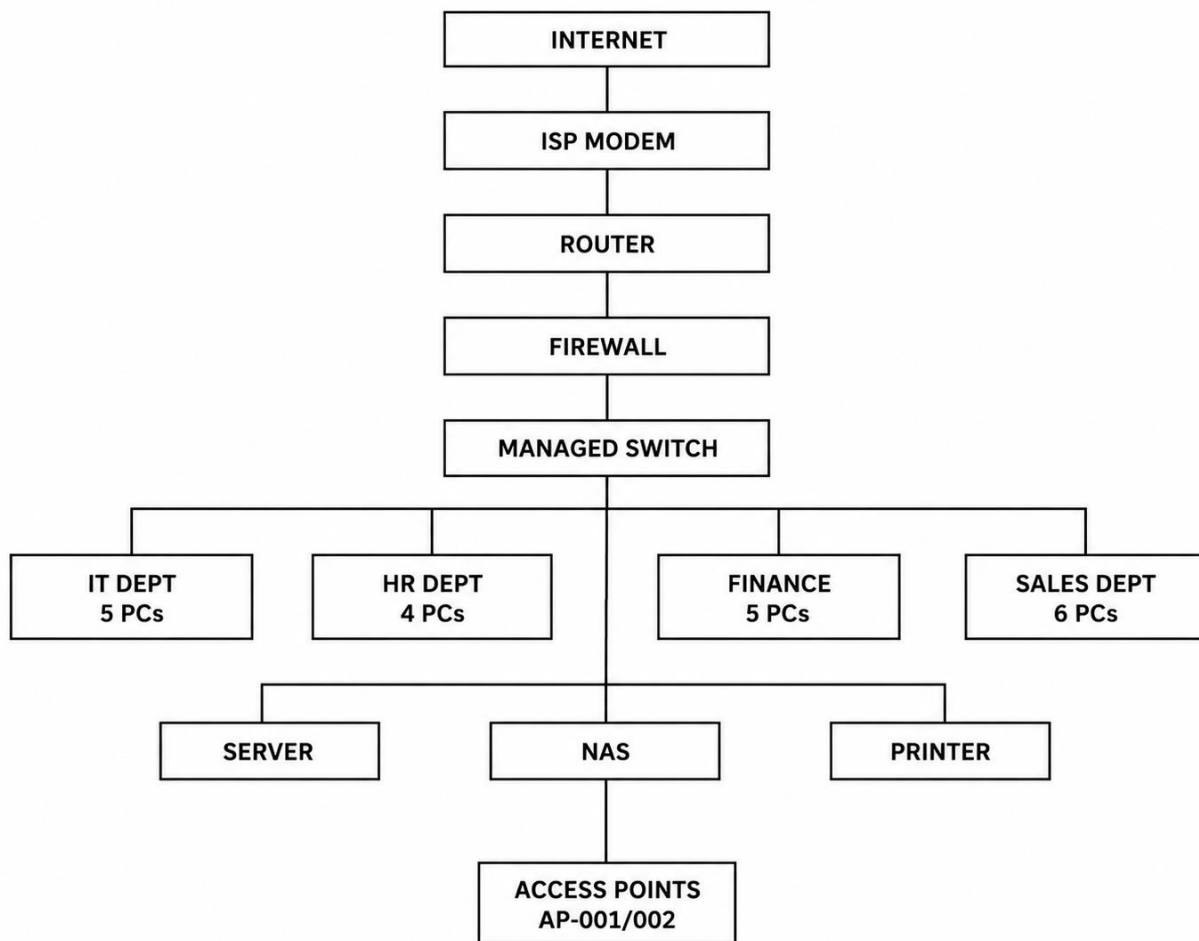
IT Department

HR Department

Finance Department

Sales Department

Recommended Topology



Export Requirements

After creating the diagram in Draw.io/diagrams.net:

Save the editable .drawio file.

Export as network-topology.png.

Export as network-topology.pdf.

Place the files inside the diagrams/ folder.

PART 6 — System Administration Roles

Helpdesk Technician

Responsibilities

A Helpdesk Technician provides first-line technical support to employees. Responsibilities include troubleshooting computer problems, installing software, configuring accounts, resolving printer issues, assisting with network connectivity problems, and documenting support tickets.

Skills

Hardware troubleshooting

Windows administration

Software installation

Customer service

Communication

Problem-solving

Ticket management

Basic networking

Common Tools

Windows Event Viewer

Task Manager

PowerShell

Command Prompt

Remote-support software

Ticketing systems

Microsoft Defender

Certifications

CompTIA A+

CompTIA Network+

Microsoft-related certifications

Network Administrator

Responsibilities

A Network Administrator maintains the organization's network infrastructure.

Responsibilities include configuring switches, routers, firewalls, wireless access points, VLANs, IP addressing, and network monitoring.

Skills

TCP/IP

DNS and DHCP

VLANs

Routing and switching

Firewall configuration

Wireless networking

Network security

Troubleshooting

Common Tools

Cisco IOS

Wireshark

Ping

Traceroute

Network monitoring software

Firewall management consoles

SSH

Certifications

CompTIA Network+

Cisco CCNA

CompTIA Security+

Linux System Administrator

Responsibilities

A Linux System Administrator manages Linux-based servers and services. Responsibilities include installing Linux, managing users and permissions, configuring services, monitoring system resources, applying updates, managing storage, and troubleshooting server problems.

Skills

Linux command line

Bash scripting

File permissions

User and group administration

Networking

Package management

System monitoring

Server security

Common Tools

SSH

Bash

systemctl

journalctl

apt

top/htop

Vim/Nano

rsync

Certifications

CompTIA Linux+

LPIC-1

Red Hat Certified System Administrator (RHCSA)

Cloud Administrator

Responsibilities

A Cloud Administrator manages cloud-based infrastructure, applications, storage, identities, networking, monitoring, and security.

Skills

Cloud computing

Virtualization

Identity and access management

Cloud networking

Cloud storage

Security

Automation

Monitoring

Common Tools

Microsoft Azure Portal

AWS Management Console

Azure CLI

AWS CLI

PowerShell

Terraform

Certifications

Microsoft Azure Administrator Associate

AWS Certified Cloud Practitioner

AWS Certified Solutions Architect – Associate

CompTIA Cloud+

Collaboration Between the Four Roles

The four professionals work together to maintain a reliable and secure IT environment. The Helpdesk Technician provides first-line support and escalates complex problems to the appropriate specialist. The Network Administrator manages connectivity, network devices, VLANs, and network security. The Linux System Administrator manages Linux servers and infrastructure services. The Cloud Administrator manages cloud-based resources, applications, storage, and identity services. Through communication, documentation, and proper escalation procedures, these professionals work together to keep the organization's IT services available and secure.

PART 7 — Infrastructure Recommendations

Internet Provider

DevCore Innovations should subscribe to a business-grade fiber internet service with approximately 300–500 Mbps starting capacity.

The service should ideally provide:

- Business technical support

- Service-level agreement

- Static/public IP option

- Reliable fiber connection

- Upgrade options

As the company expands, a secondary ISP can be added for redundancy.

Server Specifications

PART 7 — Recommended Server Specifications

Component	Recommended Specification
CPU	Intel Xeon or AMD EPYC-class processor
CPU Cores	8-12+ cores
RAM	32-64 GB ECC
OS Storage	2 × SSD in RAID 1
Data Storage	4 TB+ usable storage
Network	Dual Gigabit/2.5GbE or higher NIC
Operating System	Ubuntu Server 24.04 LTS
Power	UPS protected

The server should be located in a secure and ventilated server area.

Backup Strategy

DevCore Innovations should implement the 3-2-1 backup strategy:

3 copies of important data

2 different storage types

1 copy stored off-site

A possible structure is:

PART 7 — Backup Strategy

Backup Copy	Storage Location	Purpose
Copy 1	Primary Server Storage	Active business data
Copy 2	NAS Storage	Local backup and quick recovery
Copy 3	External / Off-site Backup	Protection against major incidents

Backups should be automated and periodically tested through restoration procedures.

Security Recommendations

The company should implement:

Business firewall

VLAN segmentation

Endpoint protection

Regular software updates

MFA

Strong passwords

Least-privilege access

Regular backups

Secure Wi-Fi

Guest network isolation

Restricted administrative access

Security awareness training

Physical server security

Antivirus

Microsoft Defender should be enabled and maintained on Windows 11 Pro workstations.

Real-time protection, security updates, and Windows updates should remain enabled.

Password Policy

Employees should use long and unique passwords or passphrases.

The company should also:

Enable MFA where available.

Never share passwords.

Use a company-approved password manager.

Disable accounts when employees leave.

Use separate administrative accounts.

Regularly review privileged accounts.

Expansion Plan

The initial infrastructure should support the current 20 employees while allowing expansion to approximately 40–50 employees.

Future improvements can include:

Additional managed switches

Additional access points

Second server

Secondary ISP

Cloud services

Centralized identity management

Enterprise endpoint management

Network monitoring

Expanded storage

Improved server-room infrastructure

Infrastructure capacity should be reviewed every 6–12 months.

PART 8 — Personal Reflection

This project taught me that system administration involves much more than repairing computers or installing software. A System Administrator must understand the organization's business requirements before designing its technology infrastructure. While working on this project, I learned how hardware, software, networking, security, backup, and documentation work together to create a reliable IT environment. I also learned the importance of creating inventories because an organization needs to know what equipment and software it owns, where those assets are located, and what purpose they serve.

The most challenging task for me was designing the enterprise network topology. I had to think about how the Internet, modem, router, firewall, switch, server, access points, printer, NAS, and different departments should connect. I also had to consider network segmentation and how VLANs could improve security. This required me to think about the network as a complete system rather than as individual devices.

Planning is important before deployment because mistakes made during the planning stage can become expensive and difficult to correct after equipment has already been purchased and installed. A proper infrastructure plan helps an organization determine what equipment is actually needed, avoid unnecessary expenses, establish security controls, plan for backups, and prepare for future growth. It also provides documentation that other IT professionals can use when maintaining or troubleshooting the environment.

This project will help me become a better System Administrator because it allowed me to practice thinking like an IT professional instead of focusing only on individual technical tasks. I learned how to connect business requirements with technical requirements and how to communicate technical decisions through tables, diagrams, and written documentation. I also realized that security and backup planning should be considered from the beginning rather than added after deployment. In the future, I can apply these planning skills when designing networks, managing servers, selecting hardware, and supporting organizations of different sizes.